

## 🎉 Faculty Achievement Spotlight 🎉

🌟 Congratulations to Our Esteemed Faculty! 🌟

### 📷 Faculty Photos:



**Name of the Faculty:** Dr.R.Venkatesan,

**Designation:** Associate Professor

**Department:** Department of Biomedical Engineering

**Achievement:** Book Chapter Publication

### 📌 Publication/Work Details

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### Chapter 10: Next-Generation IoMT Solutions for Real-Time Diabetes Monitoring and Decision Support

Sathiyamurthi K<sup>1</sup>, Venkatesan R<sup>2</sup>, Sharini B<sup>1</sup>, Ithaya Priya B<sup>2</sup>

<sup>1</sup>Department of Biomedical Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India.

<sup>2</sup>Department of Computer Science and Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India.

\*Corresponding Author mail: [venkatesan.r@egpic.org](mailto:venkatesan.r@egpic.org)

#### Abstract

The increasing prevalence of chronic diseases, particularly diabetes mellitus, has created a need for intelligent and scalable healthcare systems. The Internet of Medical Things (IoMT) enables continuous patient monitoring, real-time data collection, and improved clinical decision-making. This chapter presents a systematic review of IoMT applications for diabetes management using a multi-method approach that combines structural coding, Quantitative-based text mining, Latent Dirichlet Allocation (LDA), and the Fuzzy Analytic Hierarchy Process (FAHP). The analysis shows that wearable sensors are the core components of IoMT systems, while cloud, fog, and edge computing support efficient data processing and timely healthcare services. LDA identified major research themes related to patient monitoring and data-driven healthcare. The study highlights the importance of adopting edge-enabled IoMT architectures to improve response time, data privacy, and system efficiency. Overall, this chapter provides a clear overview of current IoMT technologies and future research directions for developing reliable, scalable, and patient-centered healthcare systems.

**Keywords:** Internet of Medical Things (IoMT), Wearable Sensors, Edge Computing, Latent Dirichlet Allocation (LDA), Fuzzy Analytic Hierarchy Process (FAHP).

#### 1. Introduction

The Internet of Things (IoT) has become an important technology in modern healthcare by connecting wearable sensors, medical devices, and mobile applications. This connectivity supports continuous health monitoring, real-time data collection, and communication between patients and healthcare providers, leading to improved disease management and personalized healthcare services [1], [2].

IoT is especially valuable in the management of chronic diseases, where regular monitoring and timely medical intervention are essential. By collecting and analysing patient data, IoT systems help healthcare professionals make informed decisions, improve treatment outcomes, and enhance the quality of life of patients [3]–[5].

The growing number of chronic diseases, particularly in developing countries, has increased the demand for efficient and affordable healthcare solutions [6]. IoT-based healthcare systems offer a practical solution by enabling remote monitoring, reducing healthcare costs, and supporting early diagnosis and timely treatment [7], [8].

This chapter provides a systematic review of IoT applications in chronic disease management, with a focus on diabetes mellitus. It discusses current technologies, research trends, and the role of edge intelligence in improving real-time decision-making, reducing latency, and enhancing data privacy in healthcare systems.

#### 2. Key Applications of IoMT in Healthcare

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Department of Computer Science, Poodi Deendayal Upadhyaya Adarsha Mahavidyalaya, Anjanga, Goolgare, Assam-78124, India

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